SOCIO-ECONOMIC TRENDS STUDY OF DISTRIBUTION OF CENTRAL HEATING AND HEALTH OUTCOMES IN ENGLAND & WALES, 2011 & 2021

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1. ABSTRACT

This study applies visual analytics methods to examine the socio-economic factors of domestic heating deprivation across 348 Local Authority Districts (LADs) in England and Wales, using 2011 Census data as the primary source with 2021 Census data incorporated for decadal comparison for 331 Local Authorities present in 2021 boundary. The absence of central heating is employed as a proxy for fuel poverty and housing inadequacy, conditions whose consequences for cold-related morbidity and excess winter mortality are well established in the public health literature (Public Health England, 2014).

In 2011, the proportion of households without central heating had a national mean of 2.54% (excluding the Isle of Scilly))and a median of 2.27% across all 347 LADs. The Isles of Scilly recorded the highest value at 26.29%; however, this represents an extreme statistical outlier attributable to its unique rural island geography and very small population, and was excluded from the Tableau visualisations to prevent distortion of the colour scale and rankings. Excluding this outlier, the effective range spans from 0.76% (Caerphilly) to 7.81% (Barrow-in-Furness). Bayesian Ridge Regression was applied both to impute residual missing values arising from geographic boundary changes between the two census periods, and to estimate standardised predictor effects with 95% credible intervals. Private rental tenure was identified as the dominant structural predictor of heating deprivation, standardised coefficient: 1.43; 95% CI [1.34, 1.52]. A Bayesian trend model extrapolating the 2011-2021 trajectory projects a national mean of approximately 0.27% by 2031, though the wide credible interval reflects substantial uncertainty over a single observed decade.

Three dimensionality reduction techniques; Principal Component Analysis (PCA), t-distributed Stochastic Neighbour Embedding (t-SNE), and Metric Multidimensional Scaling (MDS) were applied to reveal groupings of LADs with similar socio-economic and heating profiles. The resulting projections were exported to Tableau, where interactive choropleth maps, ranked bar charts, scatter plots, and tenure-stratified views were developed following Munzner's (2014) task taxonomy and the overview-first, zoom-and-filter, details-on-demand design principle. The visualisations are designed for housing policy analysts, public health researchers, and local authority planners.

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2. INTRODUCTION

2.1 BACKGROUND STATEMENT AND PROBLEM

Access to adequate domestic heating is a fundamental dimension of housing quality and a significant determinant of public health outcomes in cold counties like England and Wales. The absence of central heating serves as a practical proxy for fuel poverty defined by the UK government as a household required to spend more than 10% of its income on energy to maintain a satisfactory temperature (BEIS, 2021). Cold homes are associated with excess winter mortality, respiratory and cardiovascular disease, and deteriorating mental health, with the burden falling disproportionately on older adults, young children, and those in lower-quality private rented accommodation (Public Health England, 2014; Marmot Review Team, 2010).

Despite longstanding policy interest in fuel poverty through mechanisms such as the Energy Company Obligation (ECO), the Warm Homes Discount, and local authority Affordable Warmth schemes, the geographic distribution of heating deprivation and its structural socio-economic predictors remain incompletely understood at the Local Authority District (LAD) level. The 2011 and 2021 Censuses offer a rare opportunity to examine this question systematically, providing comparable small-area data on housing conditions, tenure, occupational class, age structure, and self-reported health across all LADs in England and Wales.

The central research question guiding this project is:

"What socio-economic factors - occupational class, housing tenure, and age predict whether a household lacks central heating across Local Authorities in England and Wales, and does the absence of heating correlate with worse self-reported health outcomes between 2011 and 2021?"

Answering this question requires integrating multiple census tables, handling geographic boundary changes between the two census periods, and communicating multivariate patterns to users with differing levels of statistical literacy.

2.2 TARGET USERS

The visualisation is explicitly designed for three distinct user groups, whose analytical needs and levels of data familiarity differ considerably. This diversity of users shaped every design decision, from the choice of chart types to the level of interactivity provided.

a. Housing and public health policymakers:
These users require spatial overviews and LAD-level rankings to prioritise intervention spending; for example, identifying which districts should receive priority Warm Homes Fund allocations or ECO scheme referrals. They benefit from choropleth maps that make geographic inequality immediately visible, ranked bar charts showing the top and bottom performing districts, and clear before and after comparisons between 2011 and 2021. They are not expected to interpret regression coefficients or dimensionality-reduced projections directly, so these are surfaced through summary tooltips rather than primary chart elements.

b. Social researchers:
This group needs to explore multivariate relationships between tenure, class, age vulnerability, and health outcomes, and is comfortable with statistical outputs. The PCA, t-SNE and MDS scatter projections are designed primarily for this audience. Scatter plots linking bad health to heating deprivation with cluster colour support hypothesis generation and pattern discovery across the full LAD distribution.

c. Local authority planners and housing officers:
This group wants LAD specific drill-down: how does their district compare to the national distribution, which tenure types are most affected locally, and how has their area changed between censuses? Interactive filters and tooltip-rich charts serve this need, enabling users to locate their specific LAD and readable values without requiring statistical expertise.

2.3 Key Questions Addressed

The dashboard is structured around 6 analytical questions:

1. How was the distribution of heating deprivation in 2011?
2. Which LADs have the highest and lowest proportions of households without central heating in 2011, and what is the geographic pattern of this inequality?
3. What socio-economic factors, private rental tenure, occupational class, and long-term unemployment reliably predict heating deprivation at the LAD level, and how certain are these estimates?
4. Does the absence of central heating correlate with worse self-reported health outcomes?
5. How has the distribution of heating deprivation changed between 2011 and 2021 across harmonised LAD boundaries?
6. Can dimensionality reduction techniques (PCA, t-SNE, MDS) reveal meaningful clusters of LADs and what do these groupings communicate about place-based inequality?

2.4 Motivation

Raw tabular census data, even when cleaned, merged, and statistically analysed cannot communicate spatial inequality, multivariate structure, or temporal change to the range of users described above. Interactive visualisation closes this gap in several ways.

First, a choropleth map makes the geographic concentration of deprivation immediately noticeable through pre-attentive colour encoding (Ware, 2020), drawing attention to high-deprivation areas without requiring users to scan hundreds of rows of data. Second, linking the map to ranked bar charts and scatter plots makes Shneiderman's (1996) information seeking more easier. Third, dimensionality reduction allows the simultaneous

visual communication of relationships across eleven variables, something no single chart or table can achieve. As Munzner (2014, p.1) notes, "the purpose of visualisation is to augment human capabilities".

The project therefore contributes both a substantive analysis of an important socio-economic inequality and a demonstration of how visual analytics methods can make high-dimensional, geographically structured census data accessible and actionable for diverse users.

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3. DATA PREPARATION

3.1 Data Sources

Six raw tables were downloaded from the Office for National Statistics (ONS) Nomis census data service (nomis web, 2011; 2021), covering Local Authority Districts (LADs) in England and Wales across two census years. All files were extracted in raw numeric format for harmonisation to allow consistent ratio calculations, normalisation, and population-weighted aggregation. The six source tables are:

2011 Census:
- Central Heating by LA.
- Central Heating by Age Group by LA.
- Central Heating by Tenure by LA – no-heat rates by owned, social, private.
- Health by NS-SeC by LA – health outcomes by occupational class.

2021 Census:
- Central Heating by LA – no-heat proportions under 2021 boundaries.
- General Health by LA – self-reported bad, fair, good, very good health.

These tables were loaded into Python using pandas and inspected for shape, column structure, and value ranges prior to any transformation.

3.2 Feature Extraction and Cleaning

Each raw file required separate extraction logic, as column positions and naming conventions differed across tables. All derived variables were expressed as percentages of the relevant denominator (total households or total population in each LAD) to ensure comparability across districts of different sizes.

The following variables were extracted and retained for analysis:

From the central heating table (2011)	- pct_no_heat_2011: % households without central heating; - pct_with_heat_2011: % households with central heating.
From the age-stratified heating table (2011)	- pct_no_heat_age_{group}_2011 for six age bands: 0-15, 16-24, 25-34, 35-49, 50-64, 65+, each calculated as no-heat count within the age band divided by total households in that age band.
From the tenure table (2011)	- pct_owned_no_heat_2011: % of owner-occupied households without heating. - pct_social_no_heat_2011: % of social rented households without heating. - pct_private_no_heat_2011: % of privately rented households without heating. - pct_owned_2011, pct_social_2011, pct_private_2011: tenure composition.
From the health x NS-SeC table (2011)	- pct_higher_mgr_2011: % in NS-SeC 1 (higher managerial/professional). - pct_routine_occ_2011: % in NS-SeC 7 (routine occupations). - pct_never_worked_2011: % in NS-SeC 8 (never worked/long-term unemployed). - pct_bad_health_2011: % reporting bad or very bad health (all classes). - pct_bad_health_routine_2011: % bad health within routine occupation class. - pct_bad_health_never_2011: % bad health within never-worked class.
From the 2021 tables	- pct_no_heat_2021: % households without central heating (2021 boundaries). - pct_bad_health_2021: % reporting bad health (2021).

Raw household and population counts were retained alongside percentages in intermediate files to support population weighted aggregation during geographic harmonisation (Section 3.3).

One notable data anomaly is the Isles of Scilly, which recorded a pct_no_heat_2011 value of 26.29% which is more than three times higher than the next highest LAD (Barrow-in-Furness, 7.81%). This extreme value reflects the island's atypical rural housing stock and very small household count (fewer than 1,000 households), making it statistically unrepresentative of any mainland pattern.

3.3 Geographic Harmonisation (2011 to 2021 Boundaries)

Between the 2011 and 2021 censuses, local government boundary reorganisations in England created 8 new Unitary Authorities by merging 25 existing 2011 districts. As a result, the 2021 census is published for 331 LADs in England and Wales, compared to 348 in 2011. To enable valid comparison, all 2011 constituent districts were aggregated into their corresponding 2021 parent authorities.

Aggregation was performed on raw household and population counts rather than percentages, because averaging percentages across districts of different sizes produces biased estimates. For example, a district of 10,000 households at 5% no-heating and one of 100,000 at 2% should aggregate to approximately 2.3%, not 3.5% (simple average). After summing raw counts across constituent districts, all percentage variables were recalculated from the aggregated numerators and denominators.

Following harmonisation, the 25 absorbed 2011 districts were removed from the master dataset and replaced by the 8 new harmonised rows, producing a final comparison dataset of 331 LADs. Two additional name corrections were applied to ensure exact string matching across files: 'Shepway' was updated to 'Folkestone and Hythe', and variant spellings of 'Rhondda Cynon Taff'. The resulting 331-LAD harmonised dataset (master_dataset.csv) is used for Bayesian modelling, and dimensionality reduction.

3.4 Missing Value Analysis and Bayesian Imputation

Following harmonisation, the 331-LAD master dataset contained no missing values in any column; all 2021 records matched successfully to their harmonised 2011 counterparts via the cleaned LAD name keys. Despite the clean final state, the decision to use Bayesian imputation was made as a principled precaution: the IterativeImputer with BayesianRidge (Pedregosa et al., 2011) was applied to the 25 feature columns, where each missing feature is estimated from all other features as predictors using a multivariate Bayesian approach.

For obtaining 348 LADs for smoother visualisation , back-harmonisation was performed using bayesian regression, as Tableau choropleth map uses 2011 boundaries. The 2021 values (pct_no_heat_2021, pct_bad_health_2021, and the change columns) are joined on via LAD name from the master dataset, allowing the 2011 map to display 2021 and change values in tooltips without altering the underlying geographic boundaries.

A second and distinct application of Bayesian imputation was required when constructing the back-harmonised hybrid dataset for Tableau. Because the hybrid dataset retains all 348 original 2011 district boundaries, the 25 constituent districts that were absorbed into the 8 new Unitary Authorities have no direct 2021 census record so their 2021 values simply do not exist as separate observations. For these 25 districts, Bayesian Ridge Regression was fitted on the 323 LADs that had complete 2021 data, using all available 2011 percentage features as predictors, and then used to predict pct_no_heat_2021, pct_bad_health_2021, and the derived change columns for each of the 25 missing districts. This allowed the hybrid dataset to carry 2021 tooltip values for every one of the 348 map boundaries without collapsing the original 2011 geography.

3.5 Data Abstraction

Following Munzner's (2014, Ch. 2) data abstraction framework, the dataset comprises the following types:
Dataset type: Table; Items: Local Authority Districts (n=331 for comparison analysis; n=347 for 2011-only analysis; n=348 including Isles of Scilly).
Attributes:
- Categorical: la_name, la_code, harmonised

- Quantitative (derived): all pct_ columns, 23 percentage variables spanning heating, tenure, age, occupation, and health.
- Ordered sequential: year (2011, 2021)

Derived attributes:

- change_no_heat_2011_2021: pct_no_heat_2021 - pct_no_heat_2011
- change_bad_health_2011_2021: pct_bad_health_2021 - pct_bad_health_2011

The key variable of interest: pct_no_heat_2011.

3.6 Standardisation and Feature Matrix for Dimensionality Reduction

Prior to dimensionality reduction, an 11-variable feature matrix was constructed from the imputed dataset using variables selected to capture the multi-dimensional structure of heating deprivation rate, vulnerability, class and share (van der Maaten and Hinton, 2008). All 11 variables were standardised to zero mean and unit variance using StandardScaler prior to PCA, t-SNE, and MDS, ensuring that variables with larger absolute ranges did not dominate variables with smaller ranges.

The final datasets exported for Tableau are:
- data_2011_348_clean.csv: 348 LADs, 2011 features only (used for 2011 maps).
- master_dataset.csv: 331 harmonised LADs, 2011 + 2021 features (comparison).
- hybrid_tableau_dataset_optimized.csv: 348 LADs, back harmonised with 2011 boundaries with 2021 values joined for tooltip display on Tableau maps.
- pca_projection.csv, tsne_projection.csv, mds_projection.csv: 331 rows each, with 2D coordinates and K-means clustering (K =4) though based on evaluation (elbow curve and silhouette score) K =2 performed best, but 4 was chosen, the next best as it is more interpretable and better for visualisation), and key contextual variables for Tableau scatter plots.

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4. TASK DEFINITION

This section defines the analytical tasks the visualisation is designed to support, using Munzner's (2014, Ch. 3) task taxonomy as a framework. Munzner decomposes tasks into three interdependent levels: Why (the user's high-level goal), What (the data target the task operates on), and How (the action performed). These are described for each of the five core analytical tasks below, which together constitute the full analytical narrative of the dashboard.

4.1 Task 1: Identify the Geographic Distribution of Heating Deprivation

Why (Goal):
Discover - the user seeks to explore which LADs suffer the highest and lowest rates of households without central heating in 2011, without prior knowledge of where inequality is concentrated. This is an open-ended spatial exploration task, appropriate for all three target user groups mentioned above.

What (Data target):
- Items: all 347 LADs (Isles of Scilly excluded)
- Attribute: pct_no_heat_2011 (quantitative, ratio scale)
- Extremes: Barrow-in-Furness (7.81%, highest); Caerphilly (0.76%, lowest)

How (Action):
Locate and identify: the choropleth map shows pct_no_heat_2011 using a diverging red-green colour scale, enabling attentive identification of high-deprivation LADs. The paired ranked bar charts (Top 15 most deprived; Bottom 10 least deprived) support precise lookup of individual LAD values and allow users to switch between extremes via a toggle filter. Tooltips provide the exact percentage on hover for any LAD on the map.

4.2 Task 2: Understand Socio-Economic Predictors of Heating Deprivation

Why (Goal):
Explain - the user seeks to understand which structural factors (tenure, occupational class) reliably predict whether a LAD has high heating deprivation, how certain those relationships are, and how different age-groups are affected. This primarily serves researchers and policy analysts.

What (Data target):

- Attributes (predictors): pct_private_no_heat_2011, pct_social_no_heat_2011, pct_routine_occ_2011, pct_never_worked_2011, pct_no_heat_age_50_64_2011, pct_no_heat_age_65plus_2011 (age vulnerability), pct_bad_health_routine_2011, pct_bad_health_never_2011.
- Attribute (outcome): pct_no_heat_2011
- Derived: standardised Bayesian Ridge coefficients with 95% credible interval (credible_intervals.csv).

How (Action):

The Bayesian coefficient plot displays standardised posterior mean coefficients as horizontal bars, with 95% credible interval whiskers. Bars coloured red indicate predictors whose CI does not cross zero i.e. reliably non-zero effects. This allow to compare effect sizes across predictors and assess statistical certainty simultaneously. The scatter plots (routine occupation x private rented x no heat, coloured by cluster) complement this by showing the bivariate relationship in the original data space.The age vulnerability bar chart (pct_no_heat by age group) supports the identify action for the 65+ group, which has the highest no-heat rate (2.80%) compared to 0-15 year olds (1.07%).

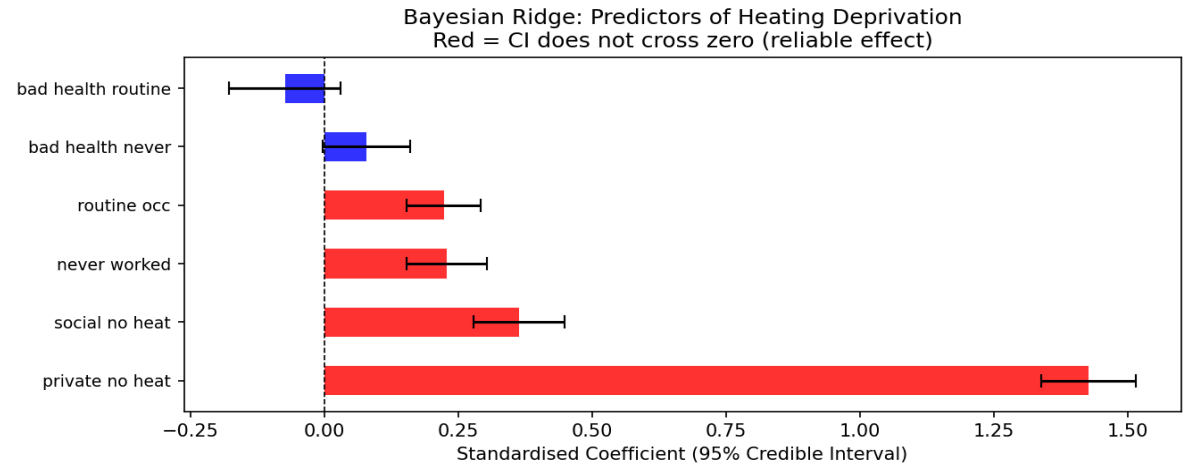


Figure 01: Bayesian Ridge coefficient plot, predictors of heating deprivation

4.3 Task 3: Explore the Relationship Between Heating and Health Outcomes

Why (Goal):

Discover and present - the user seeks to explore whether LADs with higher rates of no central heating also report worse self-reported health. This addresses the second part of the core research question.

What (Data target):

- Items: LADs (all 331; filtered to Top 15 most deprived in one view)
- Attributes: pct_no_heat_2011 (x-axis), pct_bad_health_2011 (y-axis), cluster membership (colour).
- Derived: cluster assignments.

How (Action):

Two scatter plots are used. The first is filtered to the Top 15 most deprived LADs and shows a positive trend between no-heat rate and bad health rate with a fitted trend line, supporting identification of the relationship direction. The second shows all 331 LADs coloured by cluster, revealing whether the health-heating association holds across the full distribution or is driven by specific subgroups. The Bayesian health regression coefficient plot (health_regression_coefficients.csv) provides a second model-based view, showing which predictors reliably explain bad health outcomes.

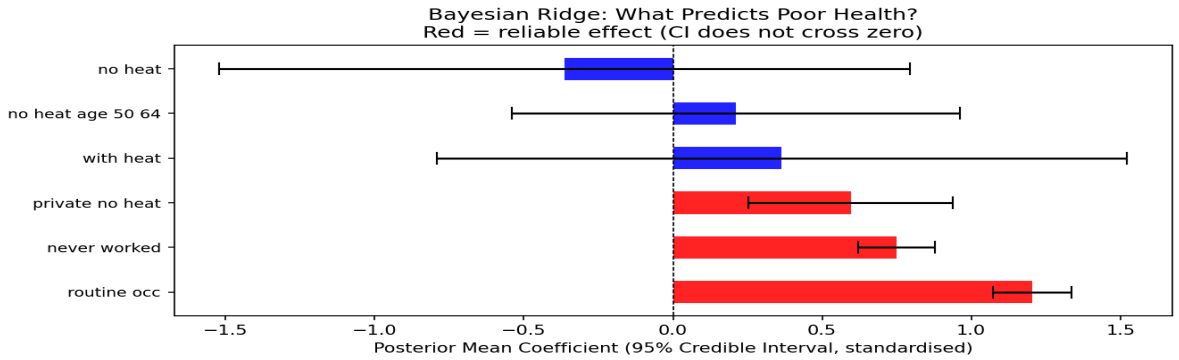


Figure 02: Bayesian Ridge coefficient plot, predictors of poor health

4.4 Task 4: Compare Change in Heating Deprivation Between 2011 and 2021

Why (Goal):
Compare - the user seeks to assess how the distribution of heating deprivation has shifted between the two census years across all LADs, and which districts have improved the most. This is the primary temporal comparison task and is relevant to all three user groups.

What (Data target):

- Items: 331 harmonised LADs (master_dataset.csv), 348 back harmonised LADs (hybrid_tableau_dataset_optimised.csv)
- Attributes: pct_no_heat_2011, pct_no_heat_2021, change_no_heat_2011_2021
- Trend: Bayesian linear model projecting mean % no heat to 2031 (trend_prediction_2031.csv: 2011=2.62%, 2021=1.44%, 2031=0.27%)

How (Action):
The change choropleth map encodes change_no_heat_2011_2021 using a sequential green scale (darker = greater improvement), enabling spatial identification of which regions improved most. The ranked bar charts for 2021 (Top 15/Bottom 10) support current deprivation levels. The stacked improvement bar chart (Top 10 LADs by improvement) shows both the change in no-heat rate and the corresponding change in bad health rate side by side, supporting joint comparison. The Bayesian trend plot (predictive distribution histogram and timeline) communicates both the projected 2031 mean and its uncertainty (95% CI: -2.64% to 3.18%), distinguishing this from a point forecast.

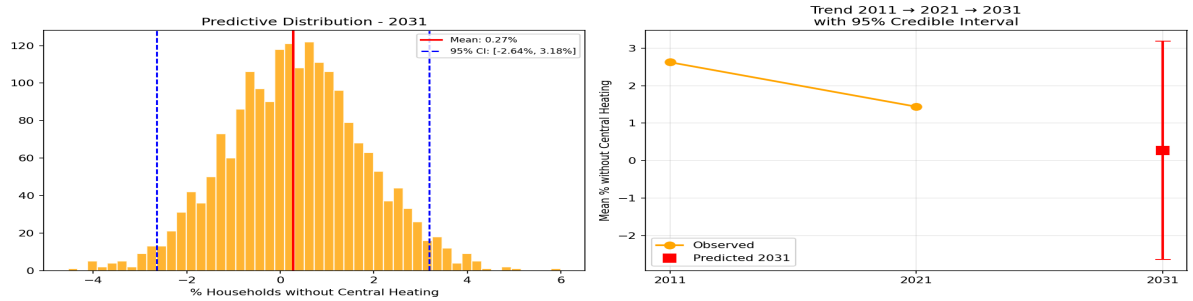


Figure 03: Predictive distribution histogram + trend timeline

4.5 Task 5: Reveal Latent Clusters of Similar Local Authorities

Why (Goal):
Discover - whether LADs naturally group into meaningful clusters based on their combined socio-economic and heating profiles, revealing place-based patterns. This is primarily a hypothesis-generation task for researchers and policymakers.

What (Data target):

- Items: 331 LADs
- Attributes: 11-variable standardised feature matrix (Section 3.6)
- Derived: 2D projections (PC1/PC2; tSNE1/tSNE2; MDS1/MDS2) and cluster assignments (4 clusters per method, from K-means on the projections)
- Cluster profiles

How (Action):

Three dimensionality reduction techniques are applied and compared:

- PCA (linear): preserves global variance structure; PC1 explains 49.3% of variance, PC2 30.2%, together capturing 79.5% in two dimensions. The scree plot and cumulative variance plot support the decision of how many components to retain.
- t-SNE (non-linear): perplexity=30, n_iter=1000, init='pca'; preserves local neighbourhood structure, revealing fine-grained cluster separation not visible in PCA.
- MDS (metric): directly preserves Euclidean distances in the original 11-dimensional space, allowing users to interpret proximity as genuine multi-dimensional similarity between LADs.

All three projections are exported to Tableau as scatter plots, coloured by cluster and with tooltips identifying each LAD and its key variable values. The use of three complementary methods allows users to assess the robustness of the cluster structure: where PCA, t-SNE, and MDS agree on groupings, the clusters are more credible (van der Maaten and Hinton, 2008; Wattenberg et al., 2016).

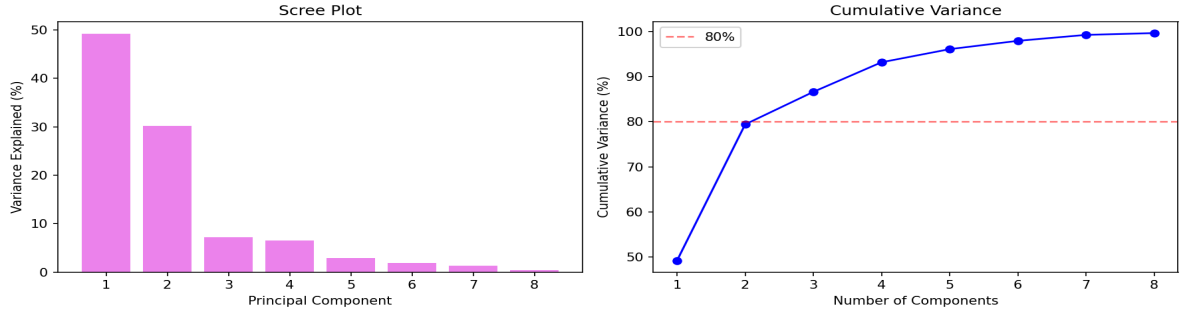


Figure 04: PCA Scree plot + cumulative variance

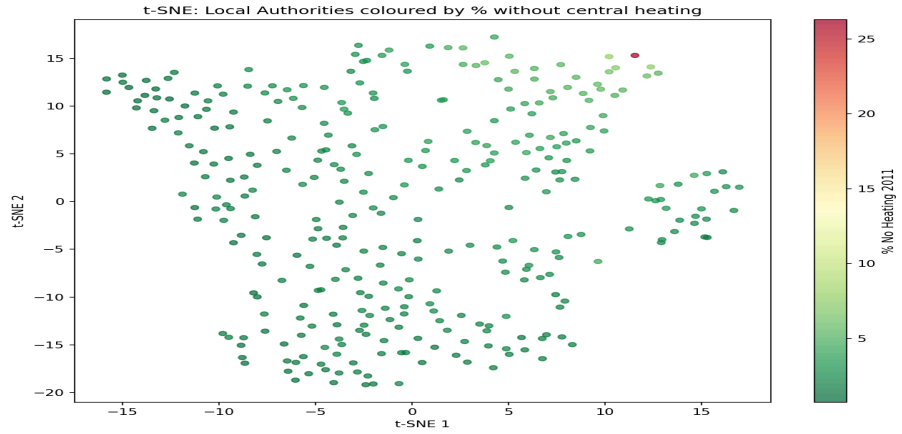


Figure 05: t-SNE scatter across 331 LADs coloured by % no central heating

5. VISUALISATION JUSTIFICATION

This section justifies every visualisation technique used in the Tableau dashboard and the Python-generated analytical plots, grounding each choice in principles of information visualisation, human perception, and the scientific literature. The justification follows Munzner's (2014) idiom design choices framework, what marks and channels are used, why they are appropriate for the data type and task.

5.1 Choropleth Maps: Geographic Distribution of Heating Deprivation

Two choropleth maps were built: one showing pct_no_heat_2011, and one showing change_no_heat_2011_2021 using a diverging red-green colour scale for both (green = low deprivation, red = high).

Justification: The choropleth is the canonical idiom for encoding a single quantitative attribute over a geographic partition (Munzner, 2014, Ch. 8). Colour hue and saturation exploit the pre-attentive processing of the human

visual system (Ware, 2020, Ch. 5), allowing users to identify high-deprivation clusters without reading individual values. This directly supports Task 1 (discover geographic distribution) and Task 4 (compare change over time).

Limitation:
The Isles of Scilly (26.29%) was excluded from both maps. Including it would have compressed the colour scale such that all other LADs appeared near-identical, rendering the map analytically useless.

5.2 Ranked Bar Charts: Top and Bottom LADs

Horizontal ranked bar charts show the Top 15 most deprived and Bottom 10 least deprived LADs by pct_no_heat_2011 in 2011, and Top 15/Bottom 10 by pct_no_heat_2021 in 2021. A dashed reference line marks the group average. A toggle button filter switches between the two views.

Justification:
Position on a common scale is the most accurate channel for communicating quantitative values (Cleveland and McGill, 1984), making bar charts superior to alternatives such as bubble charts or treemaps for precise lookup. The horizontal orientation was chosen over vertical because LAD names are long strings, which fit naturally along the y-axis without rotation. The reference line at the group mean provides an immediate anchor for judging whether a given LAD is above or below the within-group average, which is more informative than the raw value alone for policy decisions (Few, 2012). The colour coding - red bars for the most deprived, green for the least deprived, is consistent with the choropleth colour scale.

5.3 Tenure Breakdown Bar Chart: Top 5 Most Deprived LADs

A grouped bar chart shows pct_owned_no_heat, pct_social_no_heat, and pct_private_no_heat side by side for the five LADs with the highest overall heating deprivation (Barrow-in-Furness, Gwynedd, Cornwall, Torridge, Pendle). A dashed horizontal reference line marks the national average private rented with no-heat rate (most affected).

Justification:
A grouped (side-by-side) bar chart was chosen over a stacked bar chart because the analytical task requires comparing the absolute values of each tenure category across LADs. As Munzner (2014, p.96) notes, stacked bars make it difficult to judge non-base segments accurately because they do not share a common baseline. The grouped layout preserves a common zero baseline for all three tenure groups simultaneously, enabling the user to see that private rental consistently produces the highest no-heat rates, a pattern that is obscured in a stacked design. The three-colour tenure encoding (pink = owned, red = private, dark green = social) uses hue to distinguish categorical groups, which is appropriate because tenure is a nominal variable.

5.4 Age Vulnerability Bar Chart

A single bar chart shows pct_no_heat_2011 across all LADs for six age groups (0-15, 16-24, 25-34, 35-49, 50-64, 65+). Bars are colour-coded using a green-to-red sequential scale reflecting the magnitude of each bar's value (green = low, red = high).

Justification:
The sequential colour encoding on a bar chart is used here to add a redundant view of magnitude, both bar length and colour intensity communicate the same variable (Ware, 2020). Redundant view improves perceptibility and supports users who may struggle to judge small bar length differences. The chart clearly reveals a monotonic increase in heating deprivation with age for the 35+ groups (35-49: 1.84%, 50-64: 2.50%, 65+: 2.80%), which is consistent with the literature on older adults and fuel poverty (Age UK, 2022), while the 0-15 group has the lowest rate (1.07%) because child age is associated with family tenure types that have better heating provision.

5.5 Scatter Plots: Relationship Between Heating and Health

Two scatter plots were built. The first shows pct_no_heat_2011 (x) against pct_bad_health_2011 (y) for the Top 15 most deprived LADs only, with a fitted trend line. The second shows all 347 LADs coloured by cluster assignment from tableau, revealing whether the health-heating association is consistent across the full distribution.

Justification:
A scatter plot is the appropriate idiom for communicating the bivariate relationship between two quantitative variables (Munzner, 2014, Ch. 7). The trend line supports the identify task, making the positive direction of the relationship visible. The decision to show a filtered view (Top 15 only) alongside the full distribution addresses a known limitation of scatter plots: when all 348 LADs are shown, the x-axis range is compressed (most LADs cluster between 0-3% no-heat), making the relationship appear weaker than it is within the high-deprivation subgroup. The two-view design (filtered + full) gives users both perspectives without distortion (Tufte, 2001).

Colour encoding of cluster membership in the full scatter plot uses hue to overlay a categorical variable (cluster ID) onto the continuous bivariate space, revealing that the health-heating relationship is structured differently across clusters.

5.6 Dimensionality Reduction Projections

Three 2D scatter projections were produced from the 11-variable standardised feature matrix and exported to Tableau: PCA (PC1 vs PC2), t-SNE (tSNE1 vs tSNE2), and MDS (MDS1 vs MDS2). Each scatter plot colours LADs by cluster assignment from K-means (k=4) applied to the respective 2D projection. For each projection, K-means clustering (k=4) was applied to the 2D projected coordinates in Python. The value k=4 was selected after evaluating k=2,4,6,8,10 using both inertia and silhouette score on the PCA projection; k=2 produced the best silhouette score but k=4 was chosen as more interpretable given the diversity of LAD profiles.

5.6.1 PCA

The PCA scatter plot shows PC1 on the x-axis and PC2 on the y-axis, with each point representing one LAD coloured by K-means cluster (green=low cluster index, light pink=high). A dashed trend line is overlaid. The plot reveals a clear structural separation: the large light pink cluster (upper-left, negative PC1, positive PC2) groups LADs with low heating deprivation but higher health disadvantage. The dark green cluster (lower- right, positive PC1) groups the highest-deprivation LADs. PCA is justified because its explicit loadings make the axes interpretable in a way t-SNE and MDS cannot, directly communicating which original variables drive separation, satisfying the brief's requirement to communicate the original variable space to the user (Jolliffe, 2002).

5.6.2 t-SNE

The t-SNE scatter plot shows tSNE1 on the x-axis and tSNE2 on the y-axis with a dashed horizontal reference line at tSNE2=0. Points are coloured by the K-means cluster using the same green-to-red scale.

The plot reveals four visually distinct regional groupings. The dark red cluster (upper-centre) groups LADs with the highest heating deprivation. The light pink cluster (right side) forms a clearly separated group. The two green clusters (left and lower) group low-deprivation LADs.

A choropleth map ("TSNE map on % no central heating 2011") maps the same cluster assignments back onto the geographic boundaries of England and Wales. The choropleth confirms the spatial coherence of the clusters: the dark red cluster concentrates in Wales, Cornwall, and parts of northern England, exactly the same geographic pattern identified in the 2011 heating deprivation choropleth (Section 5.1). t-SNE preserves local neighbourhood structure, making it well suited to revealing fine-grained cluster separation not visible in PCA (van der Maaten and Hinton, 2008).

5.6.3 MDS

The MDS scatter plot shows MDS1 on the x-axis and MDS2 on the y-axis with a dashed trend line overlaid with the same colour scheme. MDS directly preserves Euclidean distances in the original 11-dimensional feature space, meaning proximity in the 2D plot approximates genuine multi-dimensional similarity between LADs (Kruskal and Wish, 1978). The colour encoding of change_no_heat_2011_2021 allows users to see not just which LADs are similar in their 2011 profile, but also how much each improved by 2021.

The decision to use all three DR methods rather than one reflects the recommendation that convergent cluster structures across multiple algorithms provide stronger evidence than any single method alone (Sorzano et al., 2014).

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6. EVALUATION

This section evaluates the quality of the visualisation using Munzner's (2014, Ch. 4).

6.1 Peer Evaluation

During group discussion sessions, peers were invited to observe how they could navigate the visualisation and whether they arrived at correct findings.

Dashboard 1: Peers were able to identify the geographic concentration of heating deprivation, demonstrating that the pre-attentive colour encoding was effective. The diverging red-green scale performed as intended: red clusters in Gwynedd, Cornwall, and Barrow-in-Furness were identified without any instruction. In bar chart lookup, peers successfully navigated to the Top 15 bar chart and correctly identified Barrow-in-Furness (7.81%) as the highest-ranked LAD in 2011 (excluding the Isles of Scilly). They noted that the dashed average reference line was useful for contextualising whether a specific value was meaningfully above or below the group norm.

Dashboard 2: For the tenure breakdown, peers correctly read that private rented households in Barrow-in-Furness (11.36% no-heat rate) drove the overall high deprivation rate, and that social rented households in the same LAD showed a much lower rate. This confirmed that the grouped bar design was effective for this comparison. They were also successfully able to identify the 65+ age group as the most affected by heating deprivation from the age vulnerability bar chart.

Dashboard 3: For the heat vs health scatter plot, peers were able to understand the main positive trend between heating deprivation and bad health, pointing out that the trend in the filtered view with the Top 15 LADs is considerably clearer than with all LADs. They suggested adding cluster colour-coding to the full-distribution scatter so that LADs belonging to different socio-economic groups could be distinguished within the same scatter space. This feedback was acted upon when the full scatter was updated to colour points by cluster assignment.

Dashboard 4: For change over time, peers using the change choropleth and the 2021 ranked bar chart were able to confirm that Barrow-in-Furness improved substantially between 2011 and 2021 (from 7.81% to 2.85%, a reduction of 4.96 percentage points). This cross-view navigation demonstrated that the dashboard supported multi-step analytic reasoning across linked views.

Dashboard 5: Initial PCA and t-SNE scatter plots had only a continuous colour scale with no discrete groupings, peers reported them as undifferentiated clouds of points and found it difficult to identify which LADs were meaningfully similar to one another. This feedback led directly to running K-means (k=4) in Python and adding cluster colour-coding in Tableau, which immediately resolved the interpretability problem peers had identified.

6.2 Self Evaluation

An early version of the tenure breakdown used a stacked bar chart to show the three tenure types within each LAD. Identifying the private rental bar value was difficult in the stacked design because it did not share a common baseline. This was replaced with a grouped side-by-side bar chart, consistent with Cleveland and McGill's (1984) finding that position on a common scale is the most accurate channel for quantitative judgement. The change substantially improved the legibility of the private vs social comparison.

Initially, I had only one scatter plot showing bad health versus no-heat including all 331 LADs in a single view. Because most LADs cluster between 0-3% no heat, the relationship between heating and health appeared weak when viewed at the full-distribution scale. A second plot was added showing only the Top 15 most deprived LADs, where the positive trend is considerably clearer. Having both views available made the analytical narrative more convincing than either view alone.

A further limitation I identified through self-reflection is that my analysis is cross-sectional within each census year. This means I cannot make causal claims for example, I cannot say that private rental tenure causes heating

deprivation, only that the two are reliably associated. My Bayesian regression identifies statistical associations, not causal mechanisms. I tried to be honest about this by labelling predictors as "reliable predictors" rather than "causes" in the coefficient plot, and by including the wide credible interval on the 2031 trend forecast to make clear that I am extrapolating from only two time points rather than making a confident prediction. If I were to extend this project, I would aim to use panel data across multiple census years to move closer to causal inference.

I also recognise one potential mischaracterisation in my approach. I used pct_no_heat as a binary proxy for fuel poverty, but households may lack central heating for reasons other than cost. This means my variable captures housing stock characteristics as much as household affordability. I acknowledged this in the design by including tenure and age as contextual variables, so that users could interpret a LAD's heating deprivation rate in the context of its local housing stock composition rather than treating it as a pure poverty indicator.

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7. CONCLUSION

7.1 What Was Learned About the Socio-Economic Problem

This project set out to answer one central research question: what socioeconomic factors predict whether a household lacks central heating across Local Authorities in England and Wales, and does the absence of heating correlate with worse self-reported health outcomes between 2011 and 2021? The visual analytics approach produced five substantive findings that directly answer this question.

1. Private rental tenure is the dominant structural predictor of heating deprivation, by a large margin. The tenure breakdown chart confirmed this visually: in every one of the top five most deprived LADs, privately rented households had no-heat rates above 10%, more than double the rates in owner-occupied or social rented housing in the same districts. This finding has direct policy implications: fuel poverty interventions that target social housing tenants who benefit from existing landlord obligations, may be missing the more vulnerable private rental sector, where regulatory enforcement is weaker (BEIS, 2021).

2. Older adults are the most at-risk age group. The age vulnerability bar chart revealed a clear monotonic increase in heating deprivation rates for adults aged 35 and above: 35-49 years (1.84%), 50-64 years (2.50%), and 65+ years (2.80%), compared to just 1.07% for 0-15 year olds. This age gradient is consistent with the literature on fuel poverty and cold-related mortality in older adults (Age UK, 2022; Public Health England, 2014), and suggests that age-targeted interventions such as the Winter Fuel Payment and Warm Home Discount are addressing a genuine exposure gradient.

3. Heating deprivation is associated with worse self-reported health. The scatter plots confirmed a positive trend between no-heat rate and bad health rate within the top 15 most deprived LADs, but the cluster-coloured full-distribution scatter showed this relationship is not uniform across all LADs.

4. National improvement between 2011 and 2021 was substantial but geographically uneven, and two LADs showed no improvement. Across 331 harmonised LADs, Barrow-in-Furness showed the largest mainland improvement (-4.96 percentage points), while Harlow (+0.03) and Tower Hamlets (+0.004) were the only two LADs that showed no meaningful improvement over the decade. The dimensionality reduction analysis added an additional layer of insight: PCA revealed that the 11-variable feature space is dominated by two orthogonal dimensions, a heating deprivation axis (PC1, 49.2% variance) and a deprivation-health axis (PC2, 30.2%) confirming that heating deprivation and health deprivation are related but distinct phenomena. The t-SNE and MDS cluster structures confirmed this in rural and coastal LADs with poor housing stock, and urban industrial LADs where health deprivation is driven primarily by occupational class rather than housing quality.

7.2 What Was Learned About Information Visualisation

The most important lesson was that the choice of visual encoding has consequences that extend far beyond aesthetics, switching from a stacked to a grouped bar chart for the tenure breakdown was not a minor aesthetic preference but a decision about whether users could accurately read non-base segment values. Interaction through filters, tooltips, and radio buttons also made a big difference in usability. Instead of raw comparisons, each dashboard now follows a structured path, supporting users like students and policymakers in drawing their own insights. The main analytical learning and information visualisation learning was that visualisation is to be

presented to the target user in a simplified and effective way, and to help them to understand easily what is the expected outcome.

Finally, the project reinforced Munzner's (2014, p.1) core argument that visualisation is a tool for amplifying human cognition rather than replacing it. The dashboard does not tell users what to conclude, it provides the spatial, temporal, and multivariate structure of the data in a form that allows users with different goals (policymakers, researchers, local planners) to draw their own defensible conclusions. The overview first, zoom-and-filter, details-on-demand principle (Shneiderman, 1996) proved its value in practice: peers consistently navigated from the map overview to the bar chart rankings to the scatter plots in exactly the sequence the design intended, without instruction.

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